

ENVIRONMENTAL COMPANY

DATA DRIVEN SALES OPTIMIZATION

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About

Environmental Company is a jetting and vacuum truck parts reseller and manufacturer that offers a wide range of components at competitive prices. Unlike niche suppliers that focus on limited product lines, Environmental Company provides parts across multiple industries.

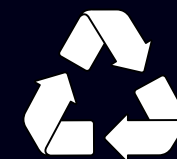
Current Customer Segments



Industrial and Commercial Contractors



**Municipalities and Public Works
Departments**



**Environmental Services and
Waste Management Companies**



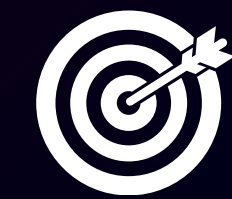
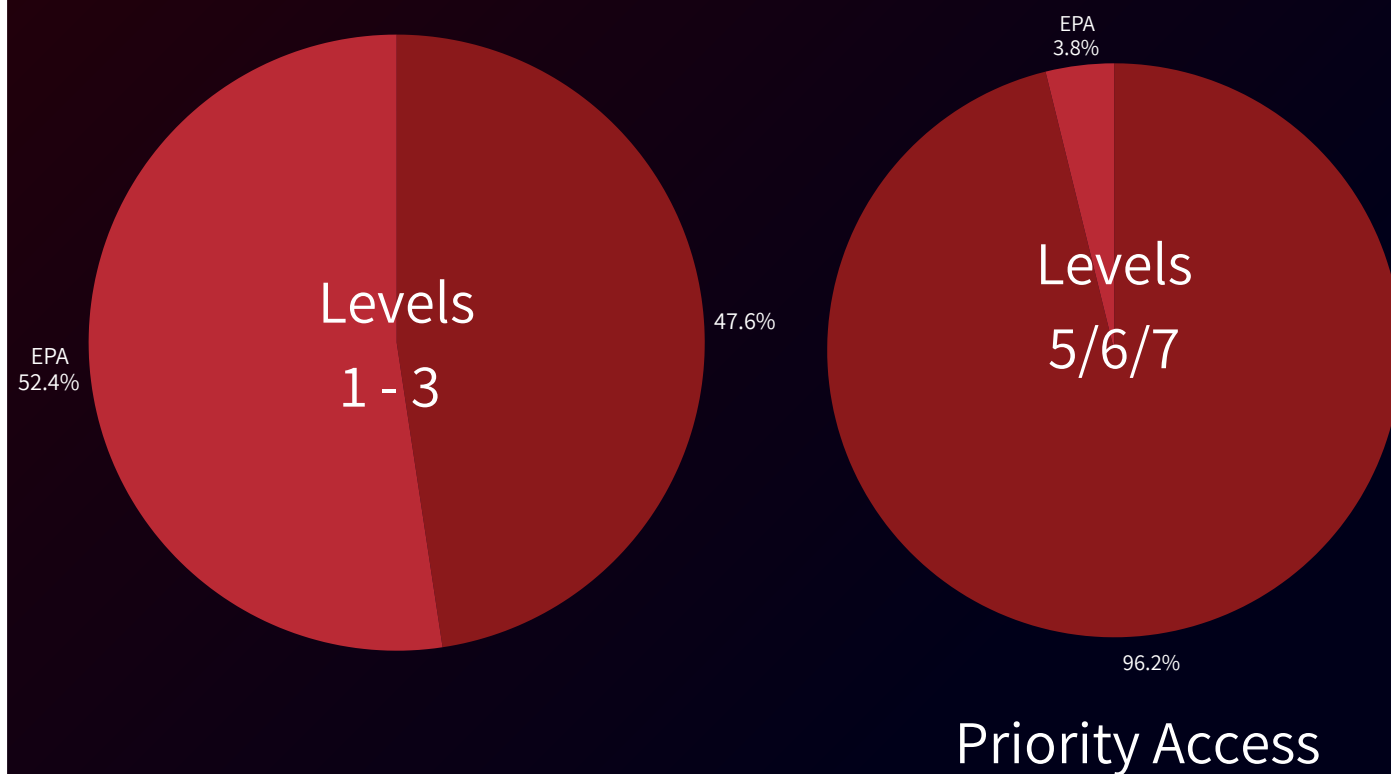
**OEMs
(Original Equipment Manufacturers)**



Equipment Dealers

MARKET SHARE

Current Market Share



GOALS

- Will a customer reorder within 30 days?
- When will they reorder?

“Reorder” or “Repurchase”: any order placed by the same customer following a prior transaction.

EXPLORATORY DATA ANALYSIS



CUSTOMER PURCHASE BEHAVIOR

- 76% of customers are classified as non-priority, contributing less to revenue, while the 24% of priority access customers generate significantly higher revenue despite being a smaller group.
- Most customers have only made a single purchase, indicating that the rare frequent buyers are extremely valuable and should be targeted for loyalty and retention strategies.
- Recommendations include retaining high spenders, upselling mid-tier customers, and identifying early loyalty signals, especially within the first 30 days after purchase to boost long-term customer value.



TIME-BASED ANALYSIS

- Environmental Company experience a strong late summer surge, steady monthly purchasing rhythms, and a critical opportunity to capitalize on mid-year growth momentum.



PRODUCT ANALYSIS

- Reveals a strong dependence on a small group of top performers, while many products in the long tail struggle to generate repeat business or significant revenue.

K-MEANS CLUSTERING

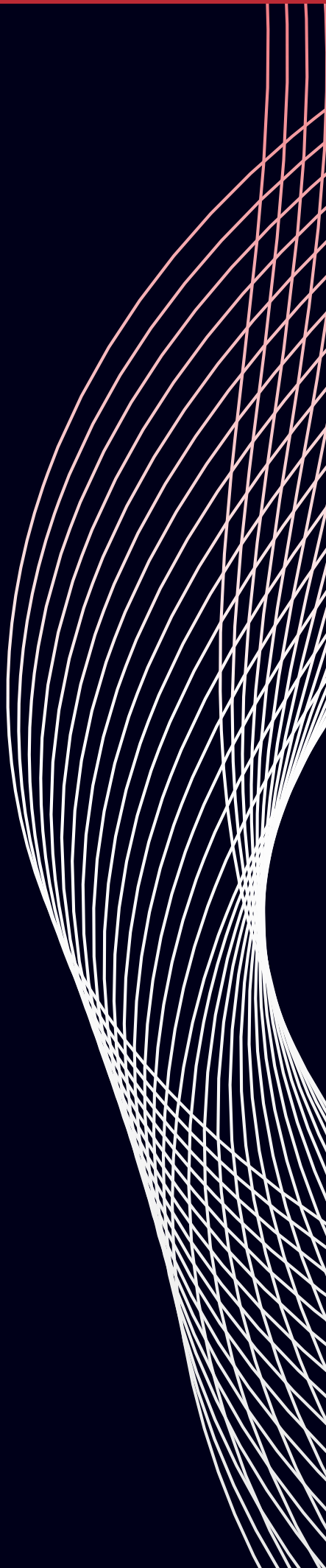
Customers were segmented into three distinct clusters using K-Means. Segmentation was based on purchase frequency, spending, and customer type. This allows to tailor marketing and retention strategies based on customer value and engagement.

What Is It?

- A method for uncovering natural groupings within a customer base based on shared behaviors or attributes
- Forms clusters of similar customers
- Identifies distinct personas
- Refine, target, and prioritize high-value segments

How Does It Work?

- Group data points into a set number of clusters based on similarity
- Goal is to minimize the distance between points in the same cluster and maximize separation between clusters.



K-MEANS CLUSTERING

Cluster 0:

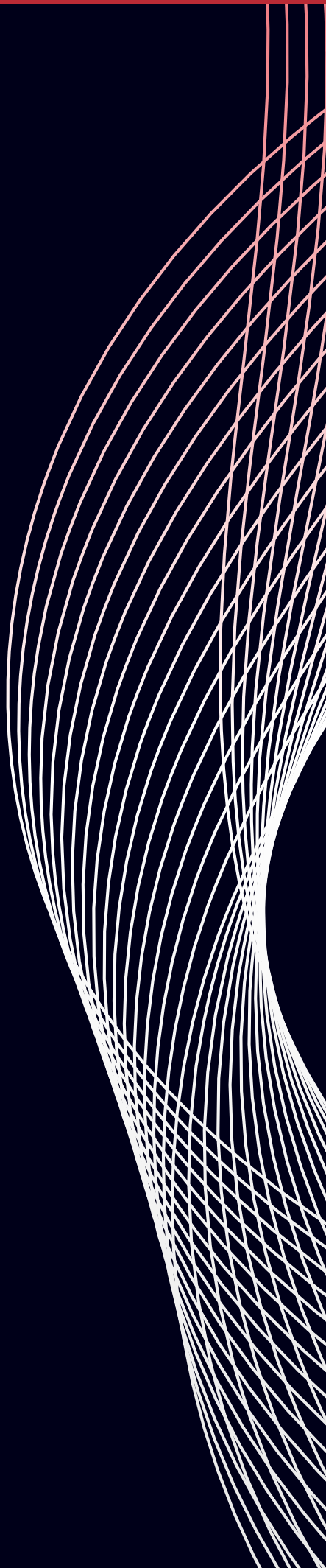
- Average of 12.6 days between orders (more spread out).
- Mostly Non-Priority - Level 1 (2, 3) customers.
- Fewer items per order and lowest total spend.
- Large concentration of customers from Texas.
- Represents newer or infrequent buyers.

Cluster 1:

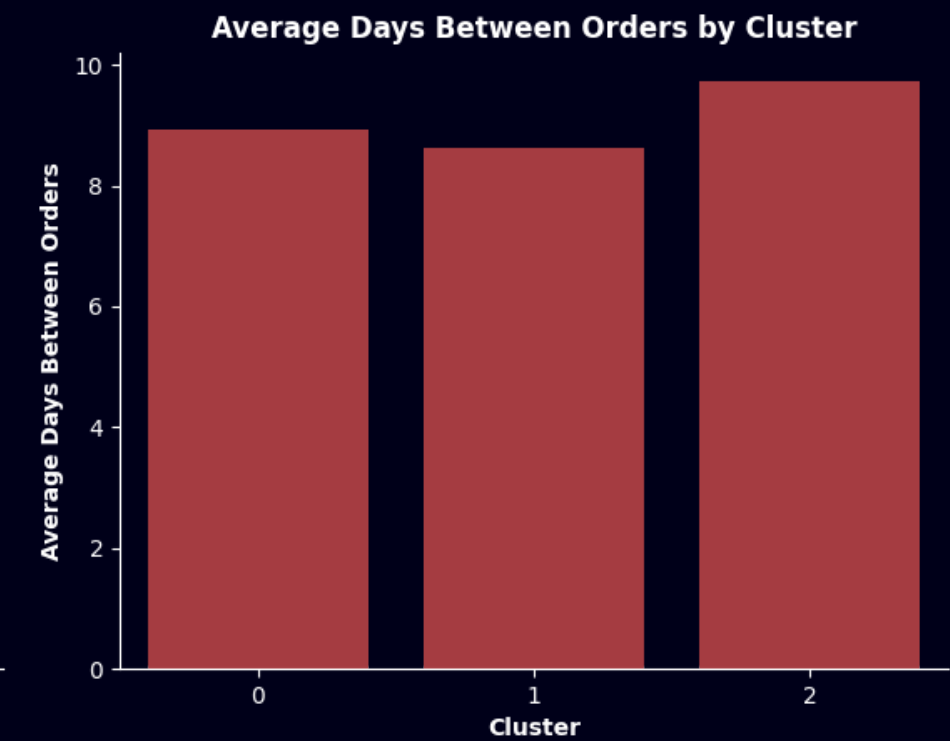
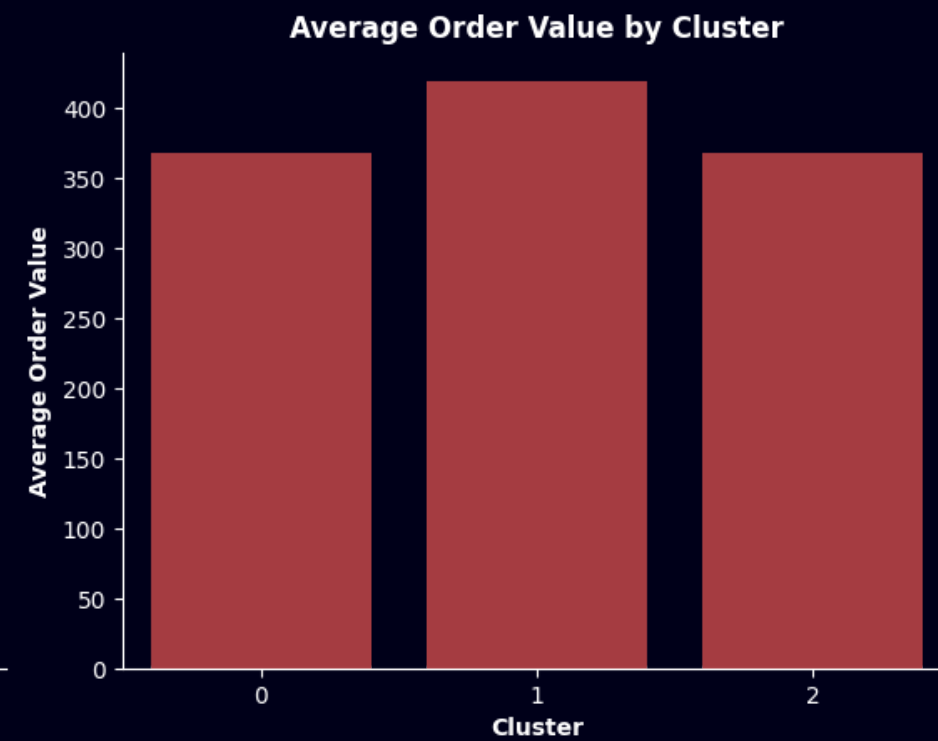
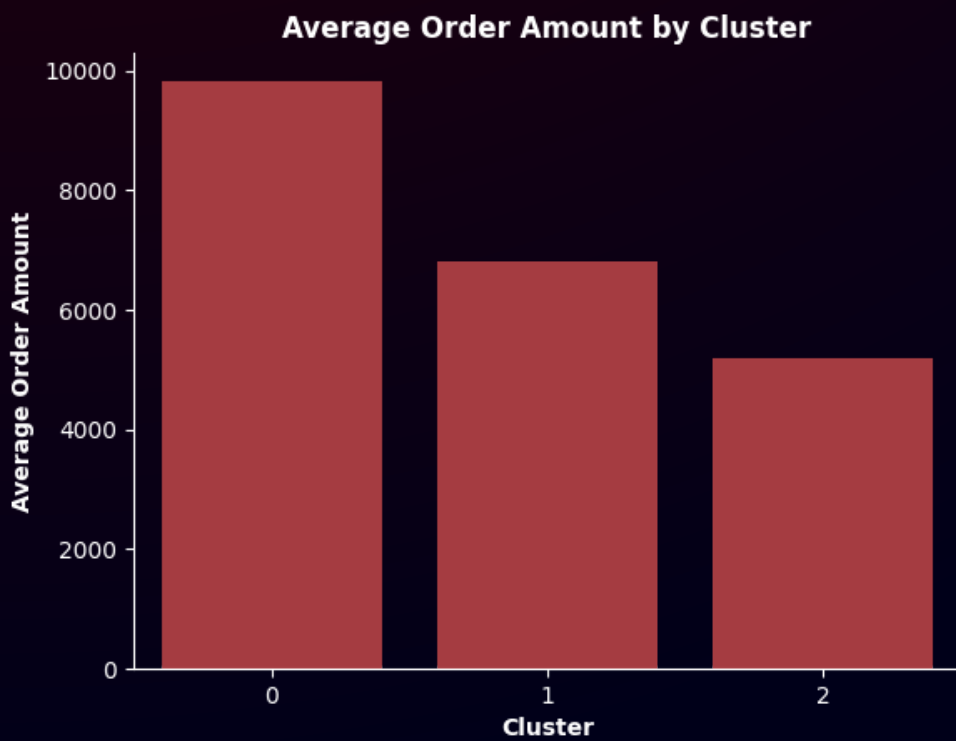
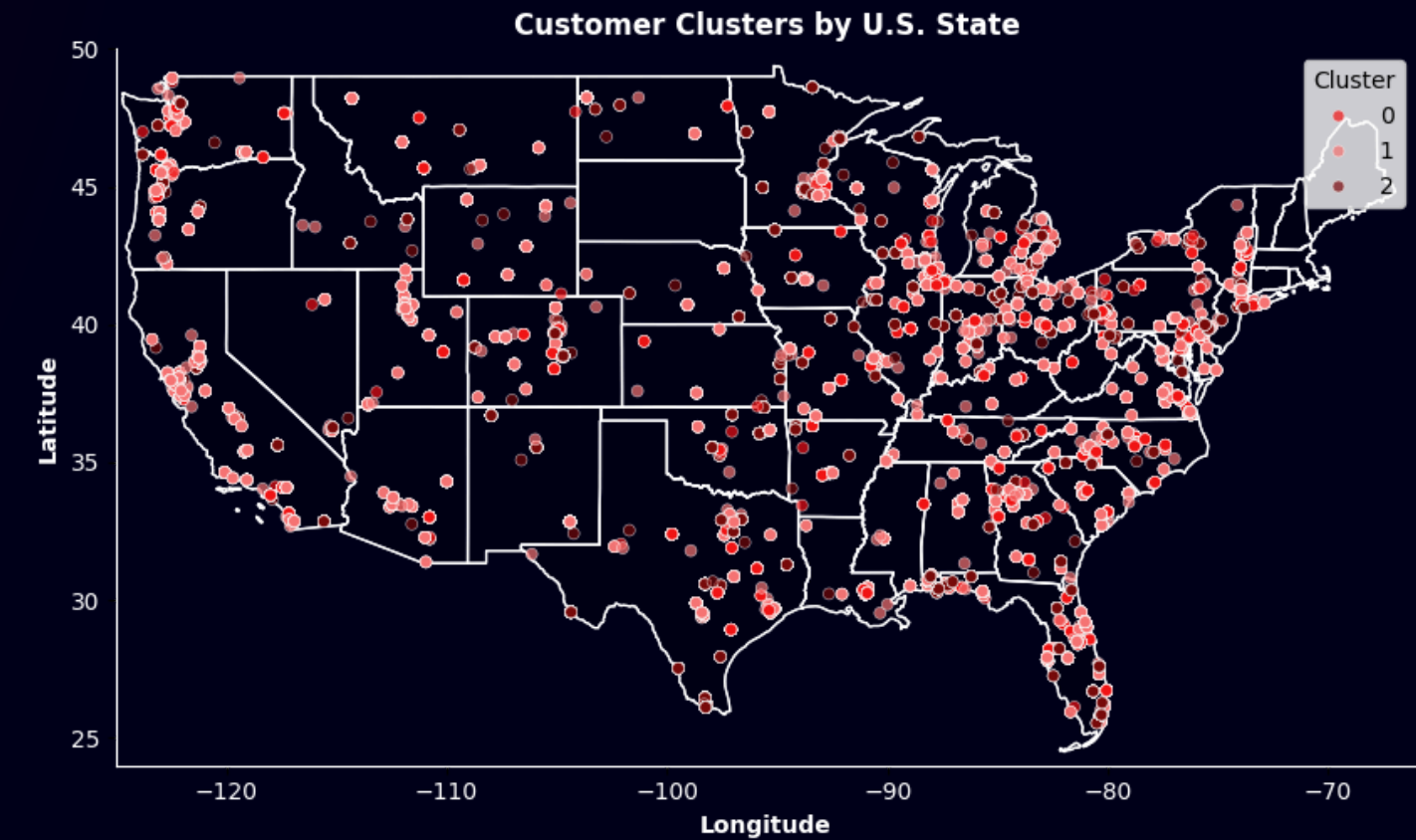
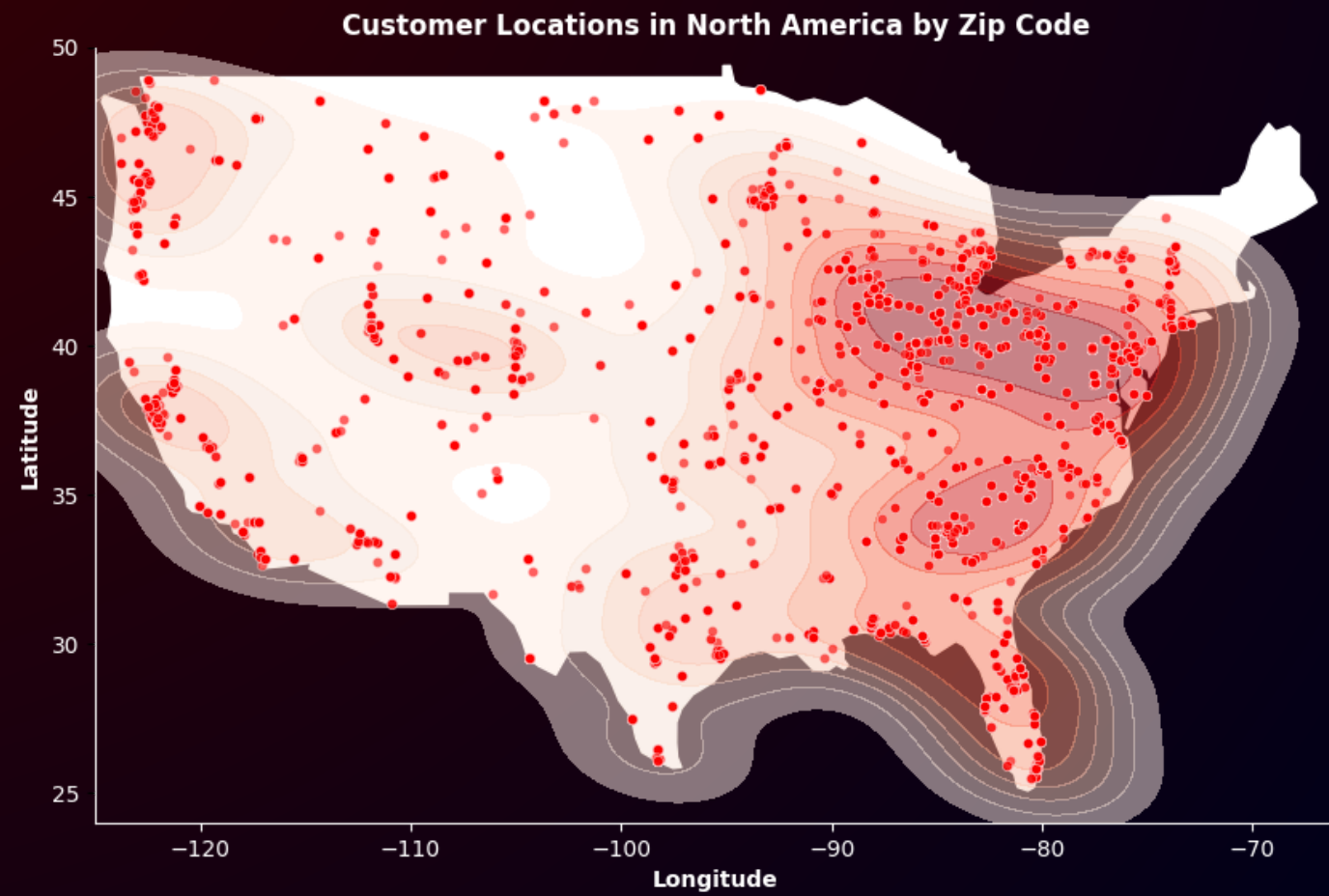
- Average of 11 days between orders.
- Primarily Priority Access- Level 5/6 customers.
- Moderate number of items per order; higher spending than Cluster 0.
- Many customers located in Virginia.
- Represents moderately engaged repeat buyers.

Cluster 2:

- Average of 10 days between orders (most frequent).
- Priority Access- Level 7 customers.
- Highest number of items per order (~4 items).
- Highest overall spending among clusters.
- Strong presence in Florida.
- Represents frequent, high-value customers.



K-MEANS CLUSTERING PT2



RANDOM FOREST MODEL

Predict reorders within 30 days

What Is It?

Multiple decision trees are used for classification and regression.

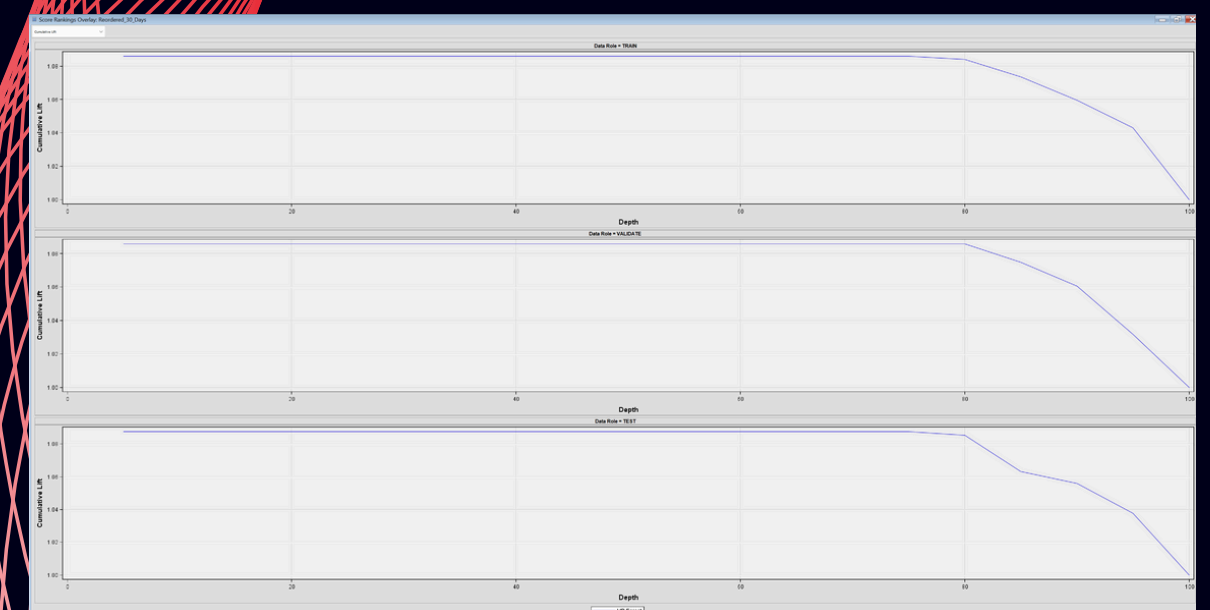
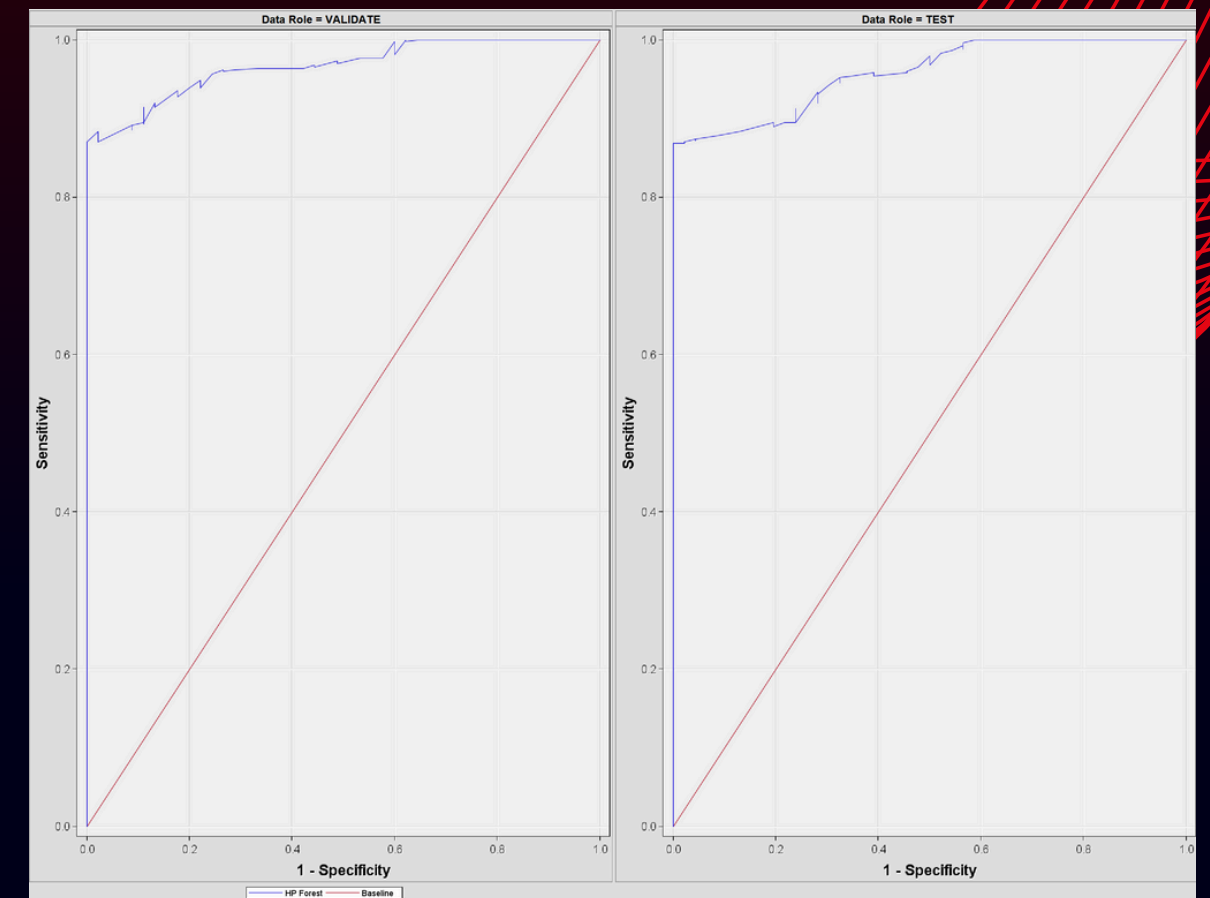
A decision tree is a flowchart-like model used to make predictions on a series of conditions.

How Does It Work?

- Starts with the important feature then base on the answer, it splits into branches
- Reach a leaf node that gives a prediction

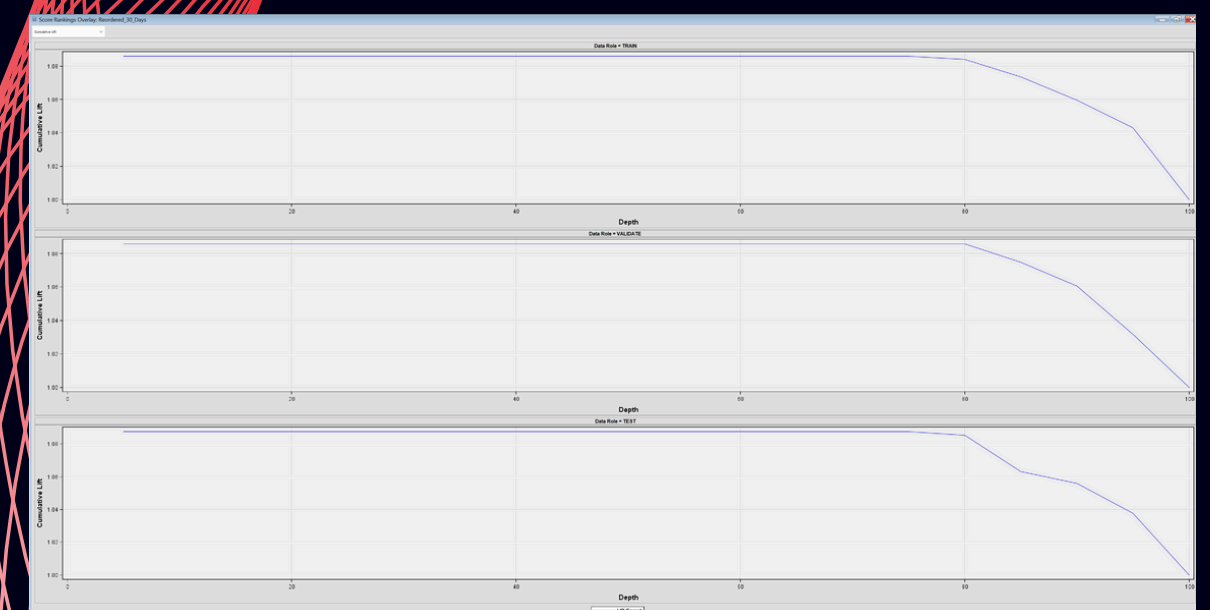
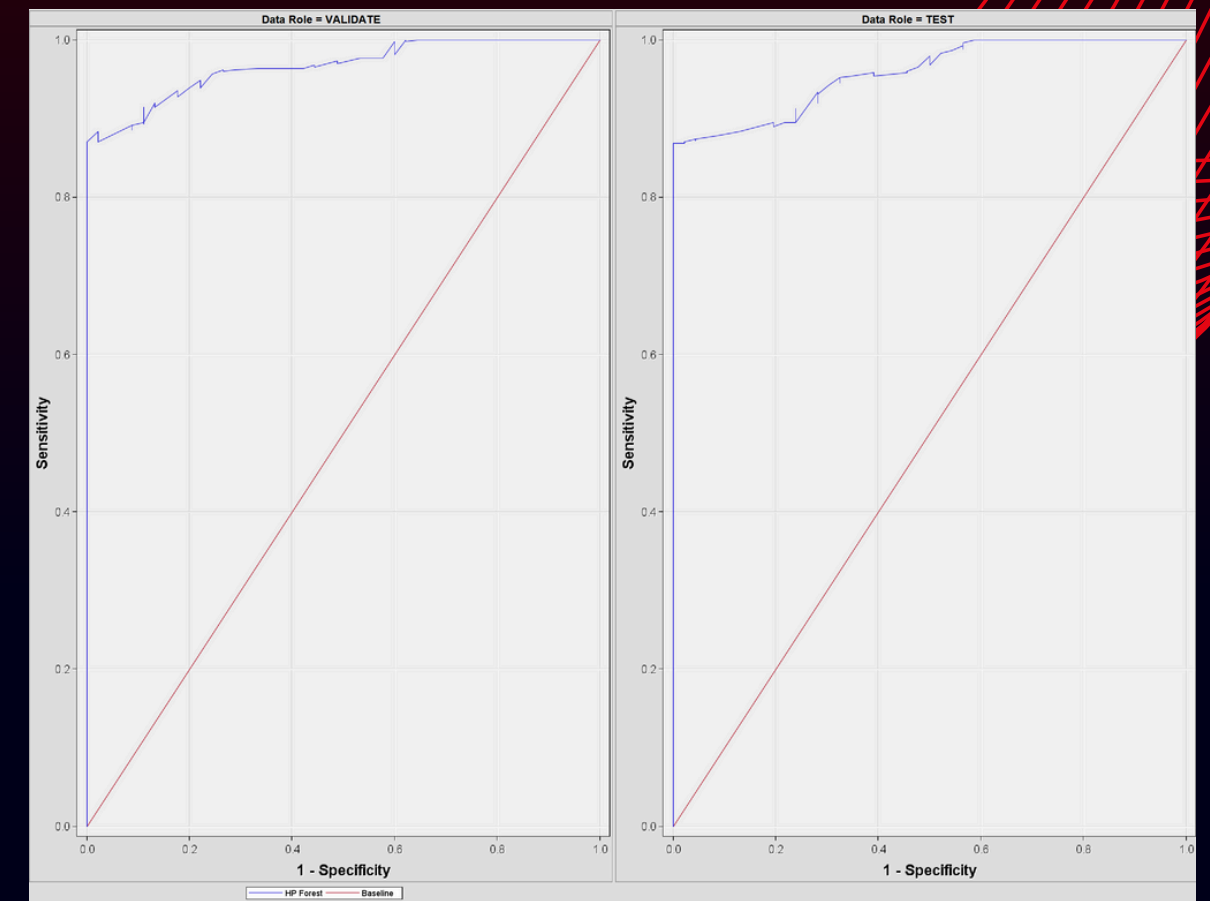
Use Cases:

- Churn prediction
- Lead scoring
- Product recommendations
- Campaign Response Predictions



RANDOM FOREST MODEL

- Goal: Predict reorders within 30 days
- Model passed all validation checks — no overfitting or unrealistic metrics.
- Strong separation between reorders and non-reorders
 - 96.6% AUC across Train, Validation, and Test
 - ~8% misclassification rate
 - RASE is low and similar across all sets (~0.22–0.23)
- Consistency across Train, Validation, and Test tells us the model is generalizing well, not just memorizing.
- Top predictors
 - Days between orders
 - Total Price
 - Customer Type.
- This model can help us proactively reach out to customers who are most likely to reorder, improving retention and reducing churn.



ASSOCIATION RULES

Part of predicting customer repurchase is to also understand what they are buying to further our strategy for cross-selling and accurate product recommendations for the website.

Conducted a market basket analysis using association rules.

What is it?

- Reveals connections between different items/elements within a data set.
- If-Then statements
- Dependency rules on co-occurrence, not causality.

How does it work?

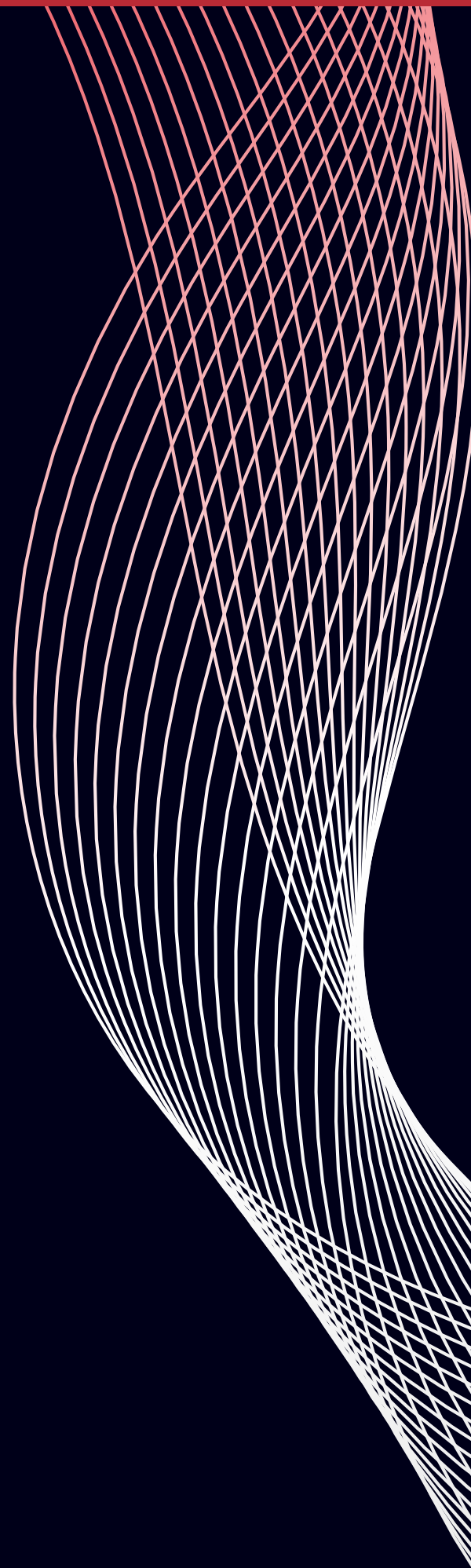
- Input transactional/behavioral data
- If-then rule
- Score with key metrics
- Filter rules
- Application

ASSOCIATION RULES

antecedents	consequents	support	confidence	lift
(vacuum truck parts, vacuum truck tubes, hydro excavation)	(vacuum truck hose fittings)	0.005686	0.666667	2.378744
(vacuum truck hose, vacuum truck tubes)	(vacuum truck hose fittings)	0.005686	0.666667	2.378744
(vacuum truck tubes, hydro excavation)	(vacuum truck hose fittings)	0.015435	0.633333	2.259807
(vacuum truck hose, vacuum truck tubes)	(vacuum truck parts)	0.005280	0.619048	2.801646
(vacuum truck parts, hydro excavation)	(vacuum truck hose fittings)	0.018684	0.589744	2.104274
(vacuum truck valves, vacuum truck parts)	(vacuum truck hose fittings)	0.007311	0.580645	2.071809
(vacuum truck tubes, manhole tools)	(vacuum truck parts)	0.007717	0.575758	2.605726
(hydroexcavation trigger)	(hydro excavation)	0.011779	0.547170	3.111160
(manhole tools, hydro excavation)	(vacuum truck hose fittings)	0.007717	0.527778	1.883172
(vacuum truck tubes, jetter hose)	(vacuum truck hose fittings)	0.007311	0.514286	1.835031

RECCOMENDATIONS

- **Focus on nurturing mid-tier customers (\$1K–\$10K) through upselling and loyalty campaigns.**
- Enhance customer profiles through improved customer tagging and firmographic enrichment for targeted marketing.
- Shift digital ad spend and outbound sales efforts to under-penetrated markets in California, New York, and Illinois, using localized case studies and testimonials.
- Integrate Random Forest and Cox model scores into the CRM to identify high-churn-risk customers for proactive outreach and prioritize top-decile leads for special treatment.
- **Set up automated touchpoints at 30, 60, and 90 days post-purchase to boost retention.**
- Improve product SEO, add "frequently purchased with" recommendations, and create product tutorials to increase repeat orders.
- Prioritize promotion of high-volume, high-margin products while avoiding overuse of discounts.
- **Integrate model scores into CRM to flag high-churn-risk customers and identify VIP prospects for targeted outreach.**



THANK YOU!

